

ULIASTAI, China

WMO: 509150

Lat: 45.5126N

Lon: 116.9630E

Elev: 2759

StdP: 13.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.4	-19.5	-30.4	1.1	-22.2	-26.6	1.4	-18.7	21.5	11.5	18.4	8.9	2.2	20	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	91.7	63.1	87.9	62.0	84.5	61.3	68.3	80.7	66.5	79.1	65.0	77.6	8.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	101.0	72.8	62.7	94.5	71.3	60.8	88.3	70.2	34.4	80.7	32.8	79.3	31.6	77.2	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	17.8	15.4	-28.1	97.1	4.4	4.5	-31.2	100.3	-33.8	102.9	-36.2	105.4	-39.4	108.7		
			-28.3	70.8	4.3	2.6	-31.4	72.6	-33.9	74.1	-36.4	75.6	-39.5	77.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	-3.5	5.5	22.0	40.8	54.7	66.2	72.0	68.0	55.9	38.5	17.6	1.5	(6)
(7)		DBStd	27.68	8.53	10.26	11.04	9.71	8.83	7.24	5.67	6.28	8.21	9.39	11.73	8.59	(7)
(8)		HDD50	7010	1659	1247	870	304	49	1	0	34	370	973	1503		(8)
(9)		HDD65	10804	2124	1667	1334	725	340	70	6	36	290	821	1423	1968	(9)
(10)		CDD50	2178	0	0	1	29	194	487	682	559	211	15	0	0	(10)
(11)		CDD65	495	0	0	0	0	19	106	223	130	17	0	0	0	(11)
(12)		CDH74	5494	0	0	0	25	384	1266	2191	1371	253	4	0	0	(12)
(13)		CDH80	1983	0	0	0	3	124	467	860	473	55	1	0	0	(13)
(14)	Wind	WSAvg	6.2	5.0	5.4	6.8	8.0	8.1	6.5	5.8	5.6	5.9	6.2	5.8	5.4	(14)
(15)	Precipitation	PrecAvg	9.8	0.1	0.1	0.2	0.3	0.7	1.6	2.8	2.3	1.1	0.4	0.1	0.1	(15)
(16)		PrecMax	16.4	0.3	0.3	0.4	1.9	2.7	3.8	6.7	7.6	3.0	1.7	0.5	0.4	(16)
(17)		PrecMin	6.0	0.0	0.0	0.0	0.0	0.0	0.4	0.8	0.6	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.5	0.1	0.1	0.1	0.4	0.6	0.8	1.3	1.5	0.6	0.4	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	39.2	59.8	77.7	89.6	94.6	97.7	95.7	85.7	71.6	52.5	32.2	(19)
MCWB			20.0	30.0	40.9	49.6	57.4	61.0	65.6	65.7	58.5	49.9	38.1	26.6	(20)	
2%		DB	19.3	33.5	51.9	70.1	83.1	89.5	92.3	89.1	80.9	65.2	44.9	23.0	(21)	
		MCWB	15.4	26.9	37.2	47.0	54.1	60.3	65.3	63.2	57.0	47.2	34.2	19.5	(22)	
5%		DB	14.8	27.8	45.7	64.6	77.8	85.3	88.4	84.9	76.4	59.5	39.7	18.7	(23)	
		MCWB	12.2	22.6	33.6	44.3	52.2	59.9	64.2	62.1	55.3	44.4	31.5	15.6	(24)	
10%		DB	10.5	22.3	40.0	59.4	72.6	81.4	84.7	81.5	72.1	54.9	34.6	14.7	(25)	
		MCWB	8.5	18.3	30.3	41.9	50.5	58.7	63.6	61.2	53.9	41.9	28.2	12.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	31.2	42.0	51.8	59.4	65.6	71.3	70.1	63.8	51.8	39.1	26.0	(27)
MCD B			24.9	37.9	57.1	73.5	79.8	81.9	83.3	86.7	76.7	68.6	50.6	32.4	(28)	
2%		WB	15.5	26.7	37.6	47.8	56.4	63.8	69.2	67.4	60.1	48.4	34.8	19.6	(29)	
		MCD B	19.0	33.2	51.4	67.7	76.3	79.4	81.1	79.4	73.8	62.2	44.0	23.1	(30)	
5%		WB	12.1	22.5	33.8	45.1	54.1	62.4	67.5	65.7	57.7	45.5	31.7	15.7	(31)	
		MCD B	14.7	27.8	45.0	62.4	73.3	76.8	80.1	76.9	71.4	58.2	39.5	18.5	(32)	
10%		WB	8.5	18.4	30.5	42.5	51.8	61.1	66.1	64.1	55.5	42.6	28.4	12.4	(33)	
		MCD B	10.5	22.3	39.8	58.8	70.4	74.8	79.3	75.7	68.9	54.1	34.5	14.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.5	22.8	23.2	25.2	25.3	22.1	20.3	21.8	24.3	22.9	19.9	17.9	(35)
MCD BR			22.7	27.9	30.9	33.3	33.5	28.2	26.2	26.8	29.9	30.0	25.1	21.5	(36)	
MCWBR			19.8	22.0	19.3	17.4	14.0	9.5	7.8	8.9	13.0	16.4	16.8	18.5	(37)	
5% WB		MCD BR	22.6	27.9	29.6	30.7	28.0	21.4	20.3	19.8	24.5	27.9	24.5	21.3	(38)	
		MCWBR	19.9	22.2	18.9	16.6	12.5	8.0	7.6	8.2	11.9	16.1	16.8	18.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.264	0.291	0.342	0.402	0.475	0.493	0.444	0.413	0.364	0.332	0.292	0.271	(40)	
taud		2.287	2.231	2.150	2.036	1.882	1.951	2.165	2.217	2.283	2.281	2.276	2.247	(41)		
Ebn at Noon		268	281	279	270	253	249	261	265	269	261	252	250	(42)		
Edh at Noon		25	32	41	50	61	57	46	42	36	31	25	23	(43)		
(44)	All-Sky Solar Radiation	RadAvg	672	1043	1472	1827	2021	1993	1953	1835	1499	1070	703	535	(44)	
(45)		RadStd	43	62	70	96	126	162	106	74	85	68	48	41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	-3.28	N/A	N/A	-1.17	N/A	N/A	N/A	N/A

Nomenclature: See separate page